LESSON PLAN

ADVANCED AIRCRAFT TECHNOLOGY (AAT) (31)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	07
Total No. of Lesson Plans	12	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Removal of loose rivets structural repair on aircraft structure
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Removal of loose rivets structural repair on aircraft structure
5. Specific Objectives	31.1.3.1 Trainee will be able to perform various repair works on aircraft structure

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Demonstration of loose rivet removal using drill and hammer; Safety precautions include PPE and fire watch; Instructor supervision required during drilling.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What are the common causes of loose rivets in aircraft structures?
(b) Why is it critical to remove loose rivets before applying new repair?
(c) List three safety precautions required before starting rivet removal.

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Identification of loose rivets • Preparation of work area • Selection of removal tools • Drilling technique for removal • Cleaning of rivet holes	L & Ds	PPT, IB	15 min
(b)	• Inspection of surrounding structure • Removal of damaged skin • Preparation of new rivets • Installation of new rivets • Final inspection and torque check	L & Ds	PPT, PDF	20 min
(c)	• Documentation of repair • Marking of repaired area • Reporting of defects • Storage of removed parts • Tool cleanup and return	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain the correct sequence for removing a loose rivet without damaging the surrounding skin.
(b) Describe the steps required to prepare a new rivet for installation after removing a loose one.

12. Home Assignment and Follow Up	Prepare a brief report on the tools and materials required for structural repair, referencing LP 02.

Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Introduction to Welding and types of welding technologies	Refer to the master syllabus reference list	LP 08

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min